

Quantifying Investor Sentiment: A Machine Learning Analysis of Bitcoin-Related Reddit Posts

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Introduction

Cryptocurrencies have rapidly reshaped financial markets. Unlike traditional assets, their valuations are highly speculative and sensitive to investor sentiment. This project evaluates whether market sentiment can be quantified by classifying Bitcoin-related Reddit posts on a 5-level sentiment scale. First, text is converted to numerical vector representations and then passed to classification models. The study tests whether simpler models can deliver sufficiently accurate sentiment predictions while reducing computational cost, and assesses how tokenization and embedding choices affect classification performance.

Methodology

The data used is the Bitcoin Reddit Sentiment dataset created by researchers from the University of Bologna (Sero, 2021), which was further processed and cleaned. The final corpus contains 35,196 posts labeled on a 5-level sentiment scale. Class imbalance was mitigated by downsampling label 1 so that labels 1 and 5 each constitute 4.197%, with labels 2, 3, and 4 at 7.20%, 2.94%, and 5.92%. Data were split into train/validation/test sets (21,117/7,039/7,040).

Embeddings:

(1) Word2Vec (Google News, 300-d): posts are tokenized with NLTK, the sentence vector is the mean of token embeddings (static, non-contextual). (2) BERT (bert-base-uncased): posts are tokenized with WordPiece, the [CLS] vector is used as the contextual sequence embedding.

Classifiers:

Embeddings were fed to two classifier families. **SVM** (*Support Vector Machine*) standardized features and used an OvR scheme across five classes. Hyperparameters spanned kernels (*linear/RBF*), $C \in \{0.1,1,10\}$, and $\gamma \in \{0.001,0.01,0.1,1\}$. **MLP** (*Multi-Layer Perceptron*) models ended with 5 output neurons and were trained with categorical cross-entropy and Adam optimizer, LR scheduling, optional dropout/BN, gradient clipping, trained up to 20 epochs with early stopping. Accuracy, macro-F1, and weighted-F1 were reported, with label-wise precision/recall/F1.

Results

- Best overall architecture: Fine-tuned BERT + MLP (5 layers, ReLU + BN), with weighted-F1 ≈ 0.85 (the most sophisticated architecture).
- Fine-tuned BERT outperforms Word2Vec pipelines across all metrics; however, if BERT is not fine-tuned (BERT + SVM), the results are consistently poor (Acc $\approx 0.44\text{--}0.45$, Macro-F1 $\approx 0.18\text{--}0.21$), indicating that contextual embeddings require domain adaptation to be effective.
- Static embeddings remain viable, with best models reaching Acc 0.77, Macro-F1 0.54 - lighter compute, but weaker on minority labels.
- Minority-class performance (robustness on labels 2–4): Fine-tuned BERT + MLP is the only family that reliably captures underrepresented classes (avg F1 ≈ 0.62 on labels 2–4).

Hyperparameter sensitivity:

- BERT + MLP: Relatively robust to head depth, metrics cluster (Acc 0.82–0.85; Macro-F1 0.63–0.73). Tuning focus: epochs/early stopping, mild regularization.
- Word2Vec + SVM: Sensitive to kernel and gamma; RBF with tuned γ performs best. Tuning focus: kernel choice, γ calibration.
- Word2Vec + MLP: Architecture-sensitive, depth and BN materially help (Macro-F1 0.29 -> 0.52). Tuning focus: depth, BN, regularization.
- BERT + SVM: Consistently poor across all hyperparameter combinations, fine-tuning is essential.

Model	Accuracy	Macro Avg F1	Weighted Avg F1
W2V_SVM_C_10_gamma_0.01_rbf	0,76	0,52	0,74
W2V_SVM_C_10_gamma_0.001_rbf	0,75	0,54	0,74
W2V_SVM_C_1_gamma_0.01_rbf	0,75	0,49	0,72
W2V_SVM_C_1_gamma_0.001_rbf	0,72	0,46	0,69
W2V_SVM_C_10_gamma_1_linear	0,69	0,48	0,68
W2V_MLP_1L_ReLU	0,68	0,29	0,62
W2V_MLP_3L_ReLU	0,7	0,3	0,64
W2V_MLP_3L_ReLU_DO30	0,71	0,31	0,65
W2V_MLP_5L_ReLU	0,7	0,3	0,65
W2V_MLP_5L_ReLU_BN2	0,74	0,49	0,73
W2V_MLP_5L_ReLU_BN4	0,77	0,52	0,75
BERT_SVM_C_1_gamma_0.001_rbf	0,45	0,19	0,41
BERT_SVM_C_10_gamma_0.001_rbf	0,44	0,21	0,41
BERT_SVM_C_10_gamma_0.01_rbf	0,44	0,2	0,39
BERT_SVM_C_1_gamma_0.01_rbf	0,45	0,19	0,39
BERT_SVM_C_0.1_gamma_0.001_rbf	0,44	0,18	0,39
BERT_MLP_1L_ReLU	0,82	0,63	0,81
BERT_MLP_3L_ReLU	0,85	0,72	0,85
BERT_MLP_3L_ReLU_BN	0,84	0,72	0,84
BERT_MLP_5L_ReLU_BN	0,85	0,73	0,85

Class	Support	W2V_SVM			W2V_MLP			BERT_SVM			BERT_MLP		
		Precision	Recall	F1-Score	Precision	Recall	F1-Score	Precision	Recall	F1-Score	Precision	Recall	F1-Score
1	2969	0.78	0.82	0.8	0.8	0.79	0.8	0.45	0.56	0.5	0.81	0.95	0.87
2	519	0.48	0.43	0.45	0.46	0.52	0.49	0.0	0.0	0.0	0.63	0.67	0.65
3	206	0.36	0.17	0.23	0.0	0.0	0.0	0.0	0.0	0.0	0.79	0.36	0.5
4	404	0.66	0.4	0.5	0.67	0.38	0.49	0.0	0.0	0.0	0.87	0.62	0.72
5	2942	0.81	0.86	0.84	0.79	0.88	0.83	0.46	0.51	0.48	0.94	0.85	0.89
Average		0.62	0.54	0.56	0.54	0.51	0.52	0.18	0.22	0.2	0.81	0.69	0.73
Weighted Average		0.75	0.77	0.76	0.74	0.76	0.75	0.38	0.45	0.41	0.86	0.85	0.85

Discussion

Investor sentiment can be quantified and may be a meaningful signal in cryptocurrency markets, serving as a practical indicator for short-horizon monitoring. Incorporating sentiment into real-time pipelines (continuous scraping and on-the-fly classification) could provide traders with dynamic dashboards and early warnings of mood shifts preceding price moves. As a modeling feature, sentiment can be fused with market microstructure signals (volume, volatility, order-book depth) to improve forecasting accuracy. Further work should test the sentiment–Bitcoin relationship, including explicit lead–lag testing. On the modeling side, domain-adapted embeddings merit attention. Training Word2Vec on crypto-specific corpora may capture jargon and neologisms better than generic vectors. For transformer routes, the weak performance of off-the-shelf BERT + SVM highlights the value of task/domain fine-tuning (e.g., fine-tuned BERT features before SVM or a fine-tuned BERT + MLP head). Key limitations to address include label noise, platform and timeframe bias, class imbalance, and concept drift in rapidly evolving crypto discourse.

References

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